

Group Work Distribution

Group Number: 5

Project: Data-Driven Retail Insights for Cost Savings Revenue Growth
(InstaCart Online Grocery Basket Analysis Dataset)

Please add each member's name and the area of the project they worked on:

Full Name	Tasks
Nathan Goyer	Optimisation of Product Mix and Predictive Recommendation Strategy through Customer Segmentation: from Exploratory Analysis to ROI Measurement (notebooks from 02 to 08) 1. Statistical Robustness & Model Generalizability I implemented a rigorous validation protocol to ensure my findings were not due to chance. I performed a stratified Train/Test split at the User ID level (80/20), ensuring that the association rules (MBA) discovered on one group of customers remained valid for an entirely unseen group. I further validated the stability of my customer segments using the Adjusted Rand Index (ARI), achieving high consistency across different data folds. 2. Behavioral Feature Engineering & Strategic Indices I moved beyond basic metrics by engineering proprietary behavioral indices. I developed the Healthy Affinity Index (HAI) to quantify a customer's nutritional orientation and the Price Sensitivity Index (PSI) to measure their reaction to cost variations. These features allowed me to transform raw transaction logs into deep psychological profiles, forming the basis for high-precision clustering. 3. High-Performance Big Data Processing with Polars To handle the scale of 33 million transaction rows, I architected the entire pipeline using Polars instead of standard Pandas. By leveraging Lazy Evaluation and optimized memory management, I significantly reduced computation time and resource consumption. This technical choice ensures

	that the analytical engine is "production-ready" and scalable for enterprise-level datasets. 4. Strategic Assortment via the Diffusion-Fidelity Matrix I categorized the product catalog using a 2D Diffusion-Fidelity Matrix based on cluster-specific quantiles. I identified products as Core (segment pillars), Niche (high-loyalty drivers), or Opportunistic (impulse buys). This allowed me to construct the "Golden Basket": a segment-specific optimized assortment that balances reach, reorder loyalty, and penetration. 5. Predictive Recommendation & Just-in-Time Triggering I built an automated engine that synchronizes product recommendations with the customer's shopping clock. By calculating the average repurchase cycle for every product, I programmed the system to trigger personalized promotions at 80% of the estimated cycle. This "just-in-time" approach captures the customer's intent precisely when they are running low on stock, maximizing conversion. 6. Financial Orchestration & Margin Protection I integrated a "Value-Aware" layer to ensure every promotion drives net profit. I modelled incremental ROI by factoring in departmental margins, sales uplift simulations, and a 0.7 cannibalization factor. I implemented a "GO/NO GO" verdict system that automatically filters out promotions that would erode margins, resulting in a validated net profit of over €5,000 for the pilot campaign. 7. Providing Result for Business analysts I acted as the core Data Architect, providing the validated datasets, financial logic, and predictive scores that fueled the Power BI executive dashboard. Without this analytical engine, the visualizations would lack the strategic depth and the 'Value-Aware' logic required for business decision-making.
Disha Srinivasa Chary	Data Enrichment and Value Engineering Lead 1. Enrichment Architecture Design

	• Designed and implemented the full data enrichment architecture, transforming raw Instacart transactions into a value-aware and health-aware analytical dataset. • Established the economic foundation enabling segmentation, association rule prioritization, revenue modelling, and promotion logic design. 2. Pricing Proxy Framework Development • Developed a structured pricing proxy framework to address the absence of transaction-level price data. • Cleaned and normalized external grocery pricing data (currency filtering, outlier removal, invalid category handling). • Computed median category-level price anchors to ensure robustness against skewed distributions and promotional noise. • Mapped Instacart departments to pricing categories and implemented fallback logic to ensure 100% price coverage. • Engineered the “unit_price” variable and constructed order-level basket value metrics supporting revenue-aware analysis and Golden Basket ranking. 3. Health Signal Integration (NutriScore) • Integrated NutriScore data from Open Food Facts with structured product matching and explicit coverage tracking. • Preserved unmatched products as “unknown” to prevent artificial imputation. • Converted NutriScore grades into numerical health indices and developed basket-level health metrics with minimum coverage thresholds to prevent distortion. 4. Validation & Methodological Controls • Performed distributional and department-level price validation to ensure realism and coherence. • Assessed NutriScore coverage and monitored missing-value behavior. • Ensured complete independence between pricing logic and health scoring to avoid circular reasoning. 5. Analytical Support & Visualization
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	- Produced enrichment-driven visualizations supporting: - Frequency vs. value differentiation, - Health–price relationship analysis, - Bundle upgrade simulations, - Segment-level Golden Basket construction.
Jérémy Trochon	Streamlit Web Application & Executive Dashboard - Designed and implemented the complete multi-page Streamlit web application architecture with navigation and state management. - Built a dual-mode interface (Executive/Analyst) enabling business stakeholders to access simplified ROI-focused views while analysts retain access to technical metrics. - Developed 5 interactive dashboard pages: Overview, Customer Segmentation, Association Rules, EDA Explorer, and Product Insights. - Implemented Polars-based data loading with lazy evaluation and caching to handle 33M+ transaction rows efficiently. - Created reusable visualization components (radar charts, heatmaps, scatter plots, bar charts) using Plotly for interactive exploration. - Integrated all notebook outputs into production-ready displays, including ROI metrics (5,378€ net profit, 174€ avoided losses), customer personas, and bundle recommendations. - Ensured data integrity by validating all displayed values against source notebooks, eliminating estimated or fabricated metrics. - Built dynamic filtering components for persona selection, product search, temporal analysis, and department-level exploration.
Armand Courvoisier	Data & Business Analyst : - Dashboard construction — Setting up all pages and visuals, applying colour coding, charters and internal best practices. - Creation of high value-added graphics — Development of visualisations to highlight important results that can be directly exploited by the business. - Business support — Presentation of key elements to facilitate the use of the dashboard and understanding of indicators. - Putting Data Science contributions into perspective — Integration and promotion of new analytical dimensions :

	○ Addition of Nutri-Score, ○ Segmentation via population clusters, ○ Demonstrate how these improvements broaden the scope of analysis compared to the old system based solely on sales.
Ndifreke Charles	Data Science & Reporting • Worked on the FP-Growth stage by helping prepare the transaction-level basket data and supporting the selection of support and confidence thresholds. I contributed to generating frequent itemsets and association rules and assisted in evaluating their strength using lift and train/test confidence comparison to ensure the patterns were stable. • Contributed to analyzing the association rule outputs, identifying meaningful product relationships and cross-selling opportunities. This involved reviewing lift values, understanding how products relate within baskets, and distinguishing frequent patterns from strategically relevant ones. • Worked on structuring and developing the final report, organizing the methodology and results sections. I helped ensure that the technical findings were clearly connected to business insights and presented in a coherent way.
Harini Magudeswaran	Data Analyst 1. Data Preparation & Modeling (Backend Work) • Created calculated columns and DAX measures • Implemented a dynamic What-If Discount Paramètres 2. Executive overview (Descriptive Analytics) on Total customers, total orders, reorder rate, etc. 3. Customer segmentation, classified customers into frequent and regular buyers and analyzed core competencies like loyalty by aisle, which helps identifying high-value customer segments. 4. Revenue and profit modeling, which answers the main issue. Simulated and calculated the line revenue, line profit same

	goes with the total revenue and profit. Analyzed revenue concentration using cumulative percentage. This provides insights into which departments and products drive profitability. 5.Promotion efficiency and ROI analysis, Implemented a dynamic discount simulation by comparing targeted discounts and untargeted discounts and calculated measurable ROI impact. This shows how targeted promotions improve financial performance.
Hassen Haj Sassi	Withdrew from the program during the project period and did not contribute to the final submission